

Bayesian ROM for uncertainty-aware forecasts of noisy dynamical systems

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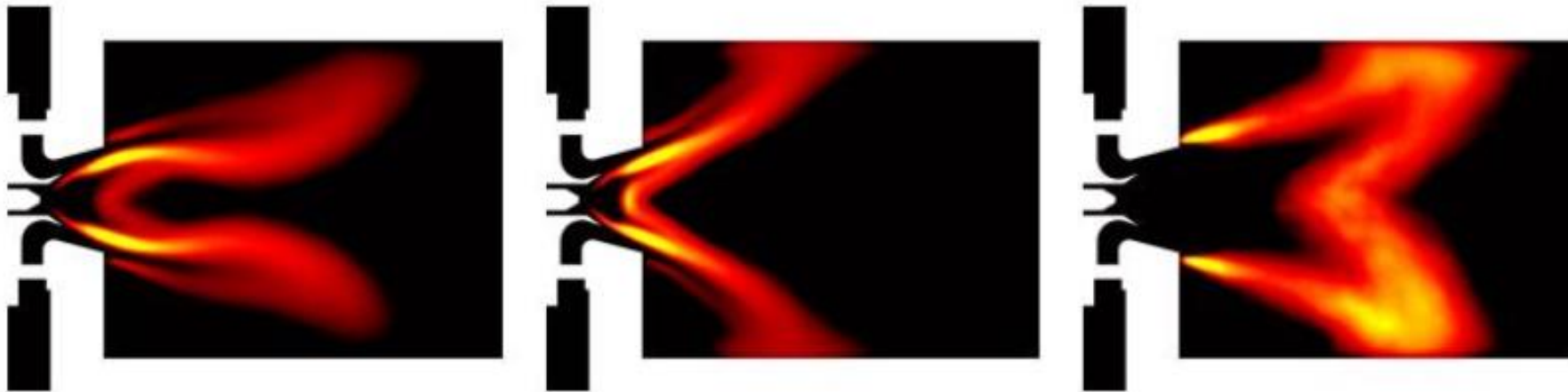
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JDSE 2026

Environmental constraints for combustion systems

Modern combustion systems face stringent **environmental and efficiency** constraints, pushing them toward lean combustion regimes to reduce thermal NO_x emissions by lowering the flame temperature. However, lean operating conditions can trigger **thermo-acoustic instabilities** and flame shape changes, causing performance degradation.



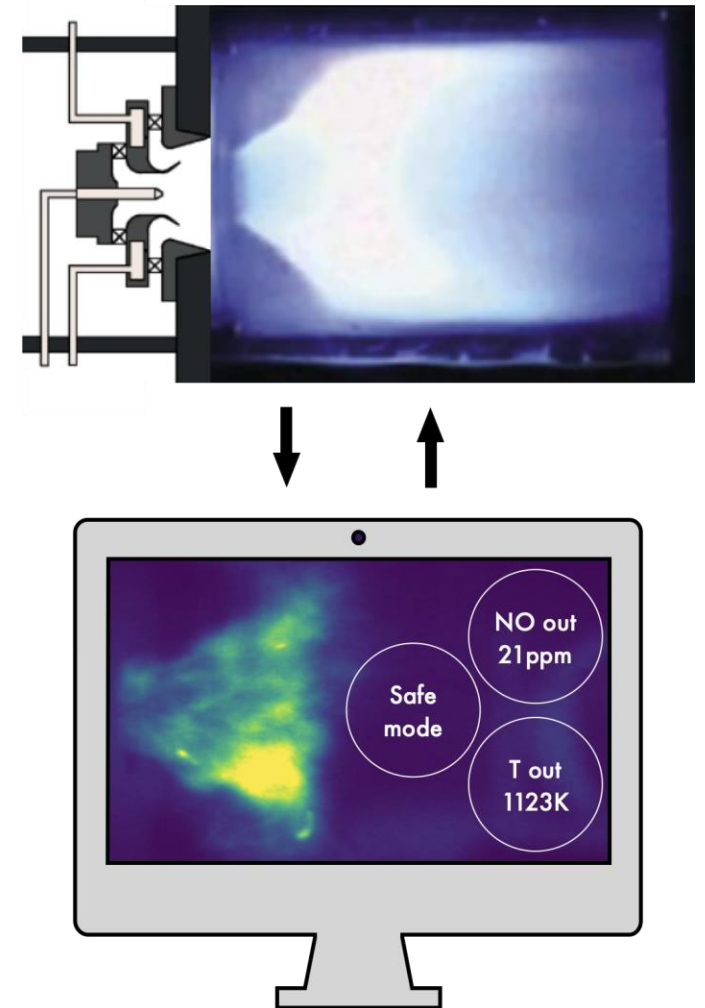
Bayesian Reduced Order Models for noisy dynamical systems

High-fidelity approaches for resolving these mechanisms are computationally expensive, whereas data-driven reduced order modelling offers a viable alternative. Digital tools are necessary for :

- speeding up design and optimization;
- preventing dangerous or inefficient conditions;
- controlling the system evolution.

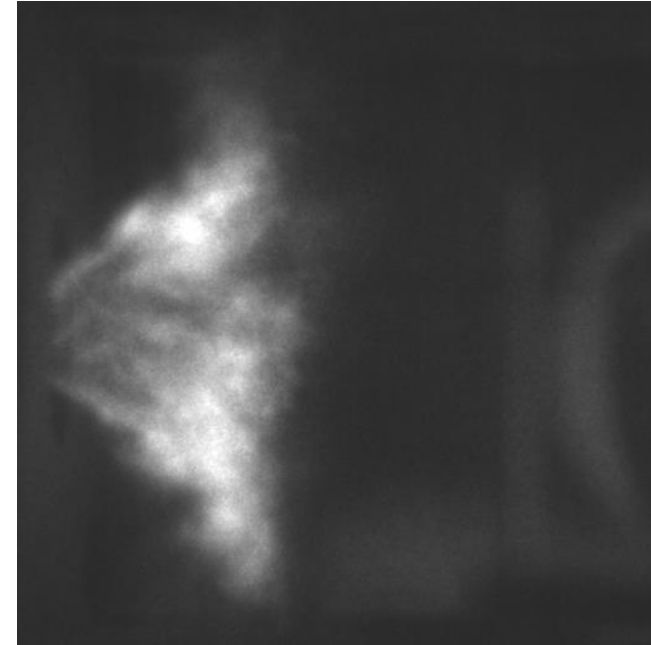
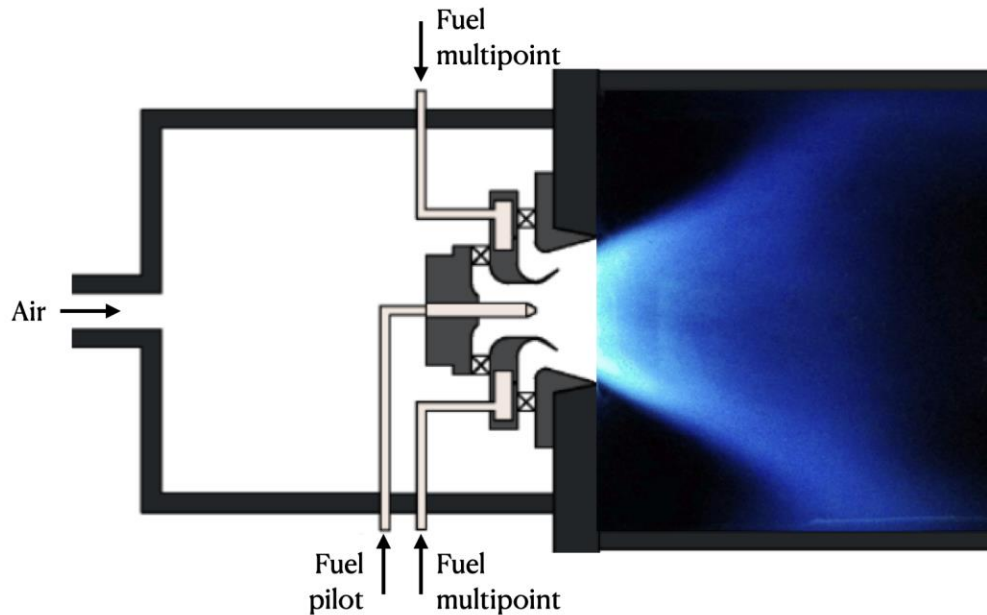
Bayesian Reduced Order Models are useful tools, capable of dealing with high-dimensional problems by:

- lowering dimensionality;
- simplifying input-output relationships;
- making uncertainty aware predictions.



Test case: experimental pulsated flame

The dataset used contains a set of chemiluminescence images from the BIMER experimental apparatus [1].



The images capture the time evolution of a swirled liquid dodecane flame, which is **pulsed** by modulating the inlet air with a **100 Hz forcing frequency**.

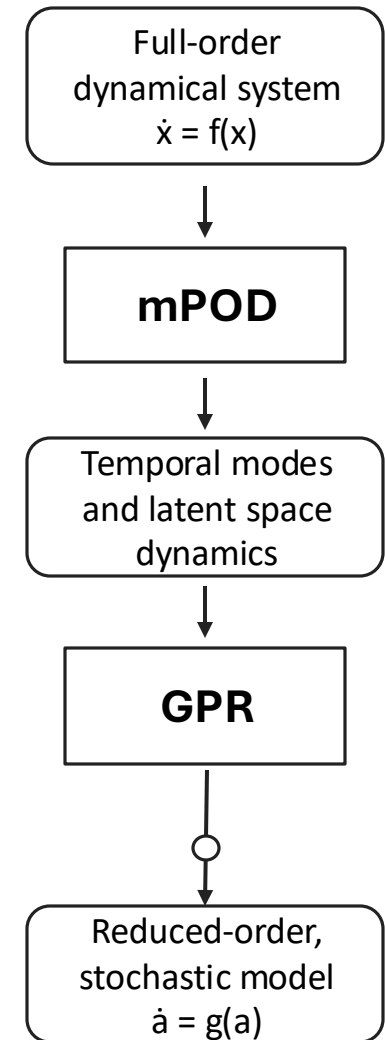
The dataset contains 5458 images with a time step equal to $\delta t = 2 \cdot 10^{-4}$ s.

Therefore, a pulsation occurs every 50 time-steps.

Previous studies and starting point

Procacci A. et al. [1] developed a ROM capable of forecasting the time evolution of the BIMER pulsed experimental flame. Their framework is based on:

- multi-scale Proper Orthogonal Decomposition (**mPOD**) for dimensionality reduction;
- Gaussian Process Regression (**GPR**), to estimate the low-dimensional trajectory from the data, and to forecast the system's evolution and its uncertainty;

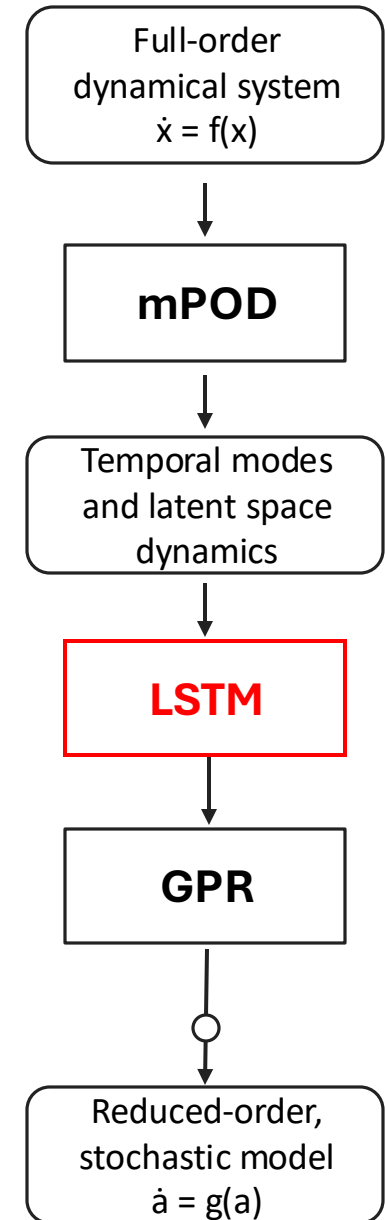


New hybrid approach: LSTM and GPR

To improve the long-term dependencies of the latent space dynamics, a new hybrid approach is developed to combine the uncertainty aware prediction of the GPR with Long-Short-Term-Memory (LSTM) capable of learning short- and long-term patterns in the dataset. [1][2]

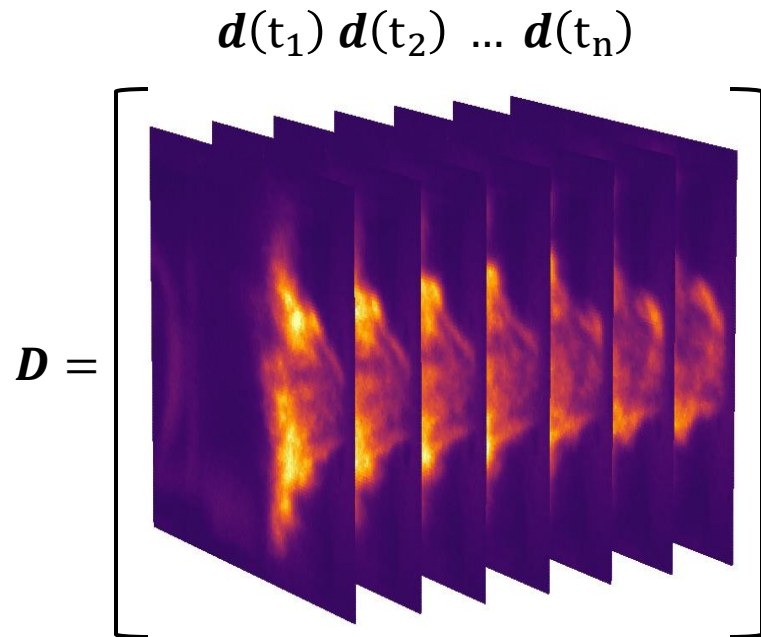
In nonlinear regression problems:

- **LSTM** models guarantee **high-accuracy** predictions;
- **GPR** models provide **interval predictions** with probabilistic significance.



mPOD: multi-scale Proper Orthogonal Decomposition

In **mPOD**[1], the encoding and decoding functions are linear transformations obtained by minimizing the reconstruction error between the original and auto-encoded datasets. After decomposing the data into **distinct frequency bands** via multi-resolution analysis, Singular Value Decomposition (SVD) is applied to extract the most energetically dominant modes.



The data matrix D , assembled by arranging n snapshots as column vectors is centered and scaled $\rightarrow D_0$

mPOD is applied:

$$K \approx \sum_{m=1}^M K_m, \quad K_m = \Psi_F (\hat{K} \odot H_m) \bar{\Psi}_F = \Psi_m \Lambda_m \Psi_m^T$$

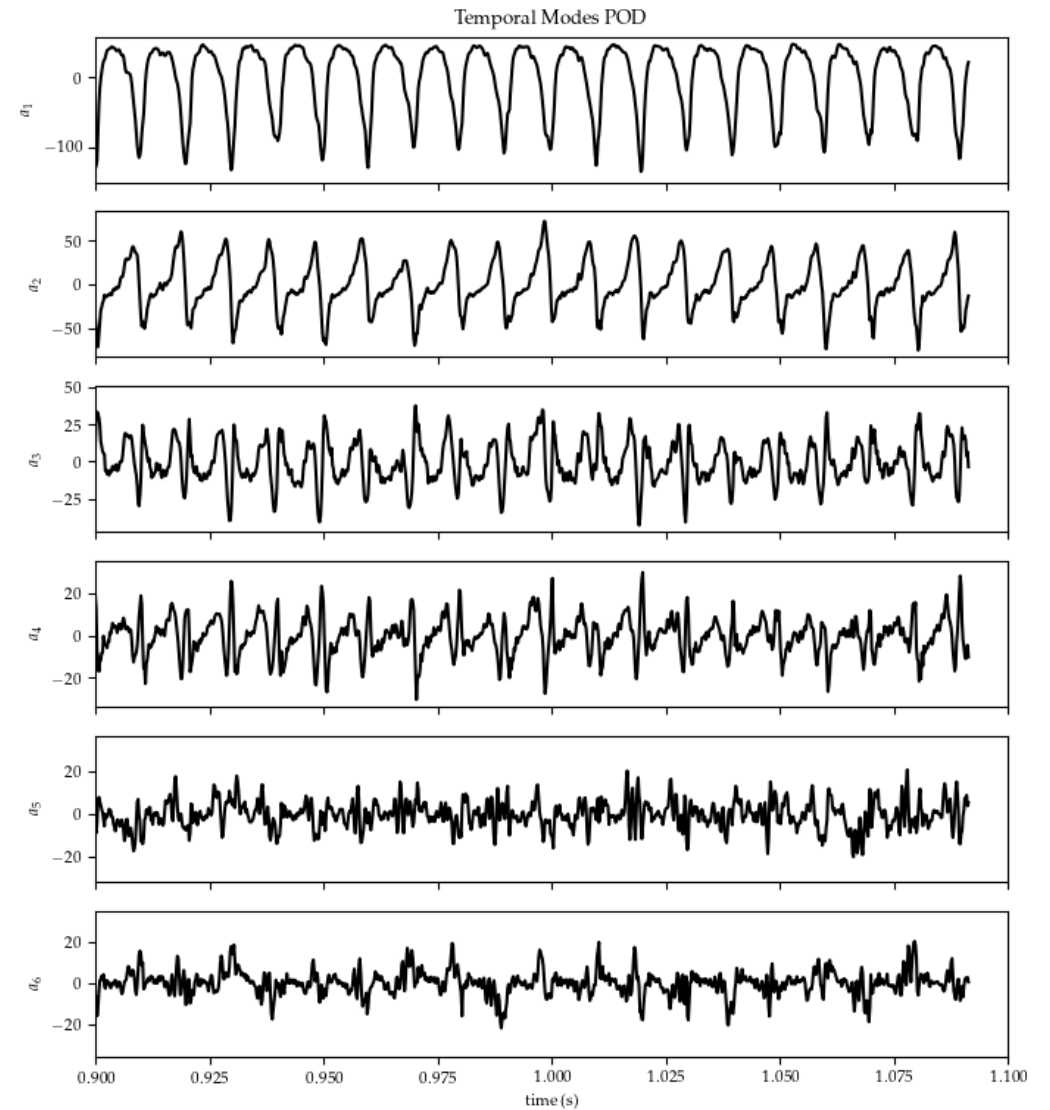
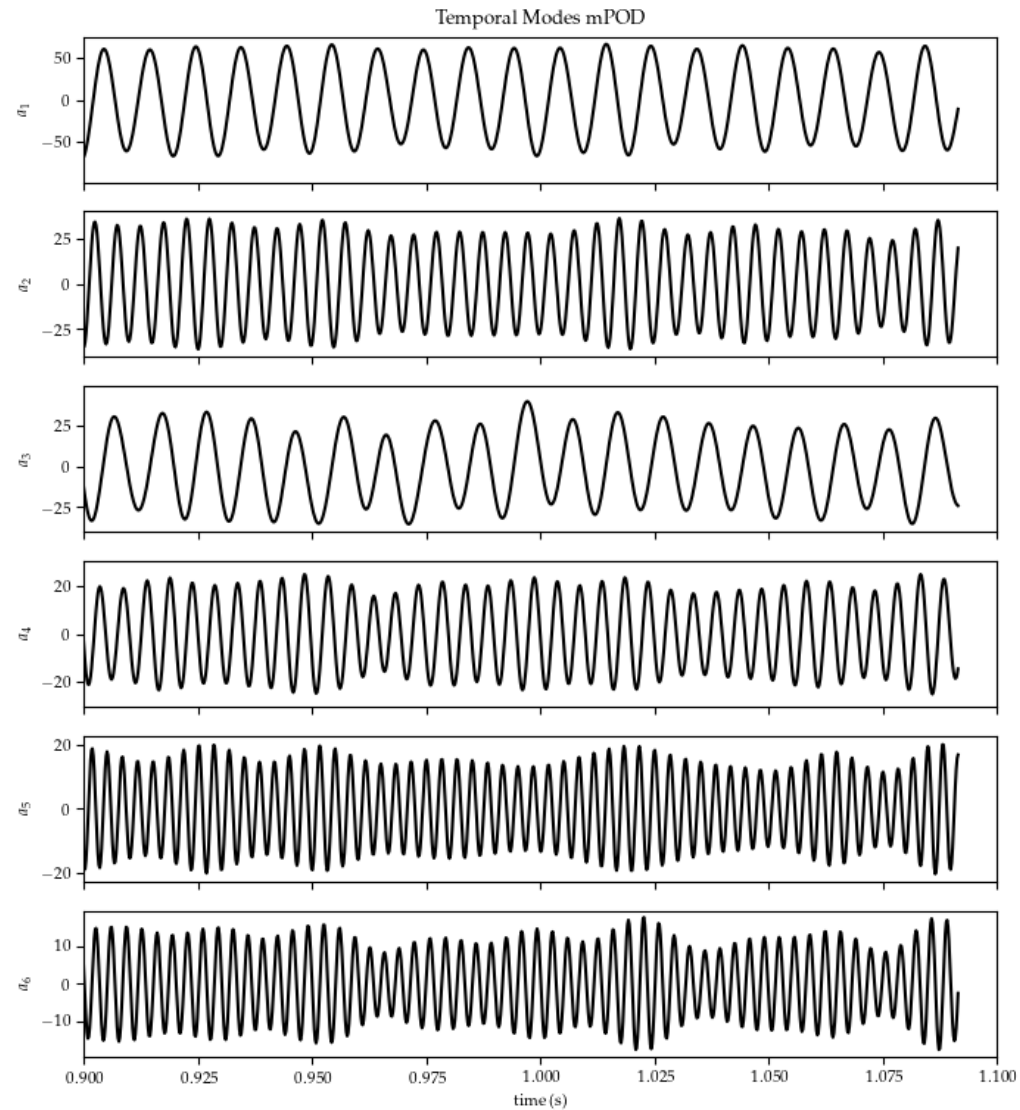
$\hat{K}$: Fourier transform of K · Ψ_F : DFT matrix · H_m : filter of band m · Ψ_m : eigenvectors of K_m

$$D_0(r, t) = \Phi(r) \Sigma \Psi(t)^T \approx \Phi(r) A(t)$$

The first **6 principal modes** have been retained from this process, accounting for approximately 79% of the total variance of the system.

$$A(t) = \begin{bmatrix} a_1(t_1), a_1(t_2), \dots, a_1(t_n) \\ a_2(t_1), a_2(t_2), \dots, a_2(t_n) \\ \dots & \dots & \dots \\ a_6(t_1), a_6(t_2), \dots, a_6(t_n) \end{bmatrix}$$

Comparison between mPOD and POD temporal modes



LSTM – Long-Short Term Memory Networks

LSTMs are a kind of Recurrent Neural Network (RNN) designed to learn long-term dependencies of time series. Inside an LSTM cell, highly nonlinear activation functions are found, making it capable of dealing with complex dynamics.

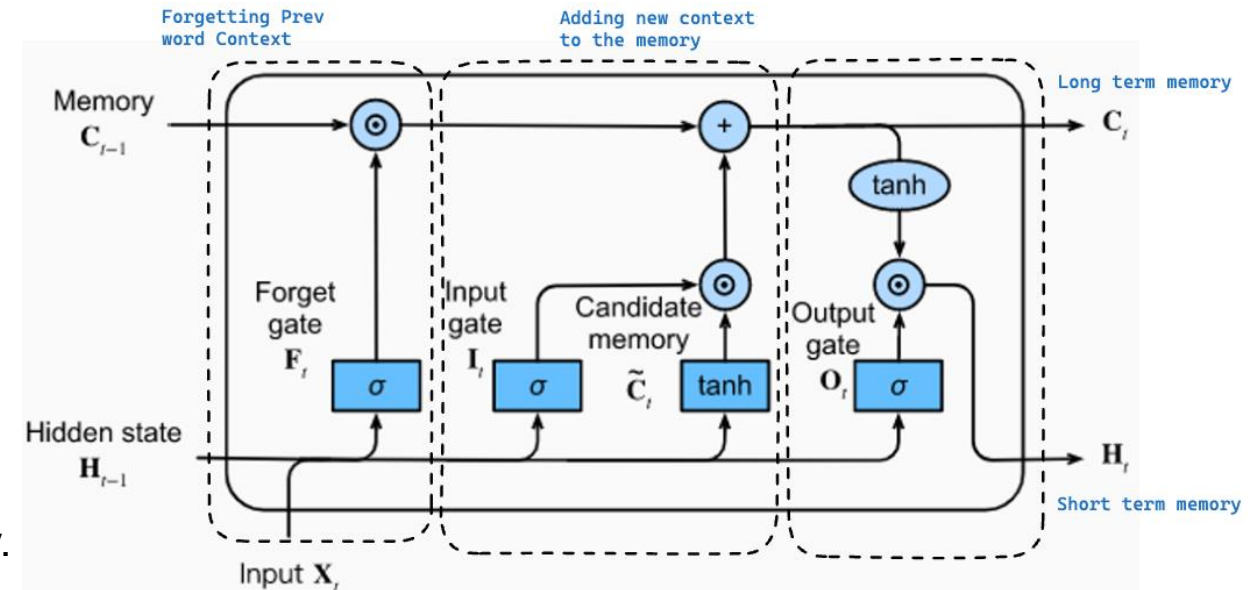
They are characterized by two internal states propagating through entire network chain :

- a **cell state** (C_t), responsible for the long-term memory;
- a **hidden state** (H_t), responsible for the short-term memory.

Training logic:

- Input: $X_j = a_i(t_j)$
- LSTM update: $H_j, C_j = LSTM(X_j, H_{j-1}, C_{j-1})$
- Output prediction: $\widehat{a_i}(t_{j+1}) = f_{out}(H_j)$

with $i = 1, 2, 3, 4$ and $j = 1, \dots, n$

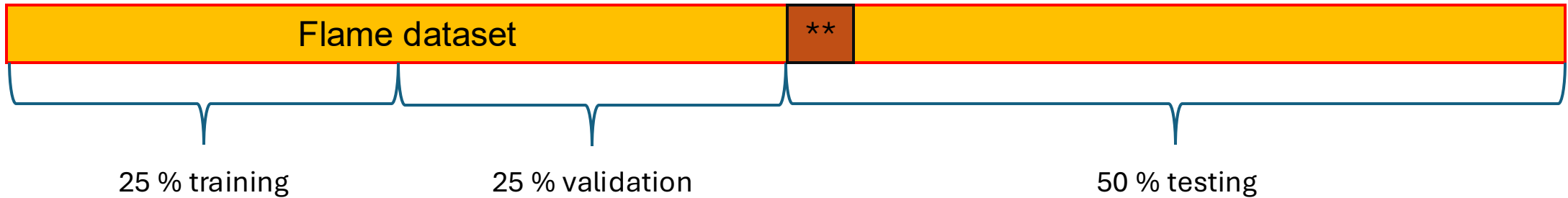


Source: Medium (@robin5002234), "LSTM RNN: An In-Depth Look at its Architecture"

Goal: learning the low-dimensional mapping function g that evolves the system from one snapshot to the next:

$$a(t + \delta t) = g(a(t), t)$$

Management of the dataset for LSTM training



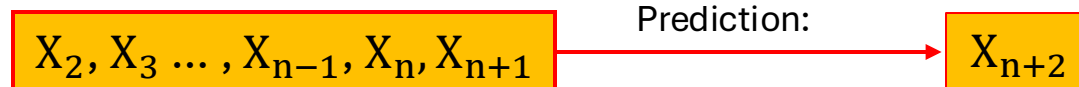
Sliding window approach:

First n points are chosen to be part of a sequence that is fed as input to the the LSTM in batches during training:



where $X_j = a_i(t_j)$.

Next sequence:



And so on until the last available point...

** : The first X_n points are part of the first sequence and therefore are treated like an “initial condition” of the predictions. Considering as a first sequence the last X_n point of the validation set would cause data leakage.

GPR – Gaussian Process Regression

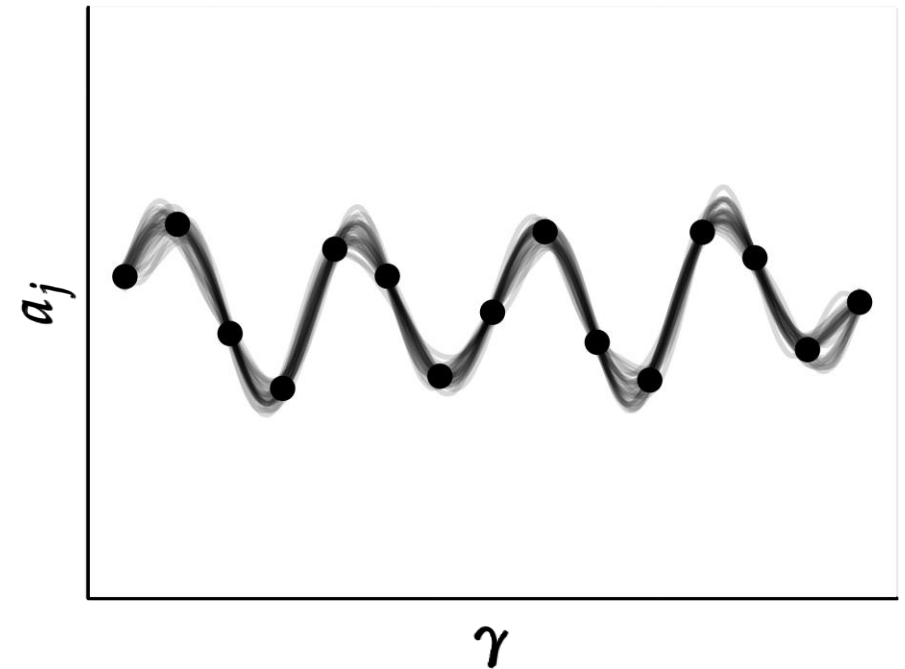
In this framework, Gaussian process regression (GPR) is used to enhance the prediction capabilities of the LSTM in the probabilistic domain.

We can model the function $g(\mathbf{a}(t), t)$ as a **Gaussian Process (GP)**, i.e., a **normal distribution of random functions**:

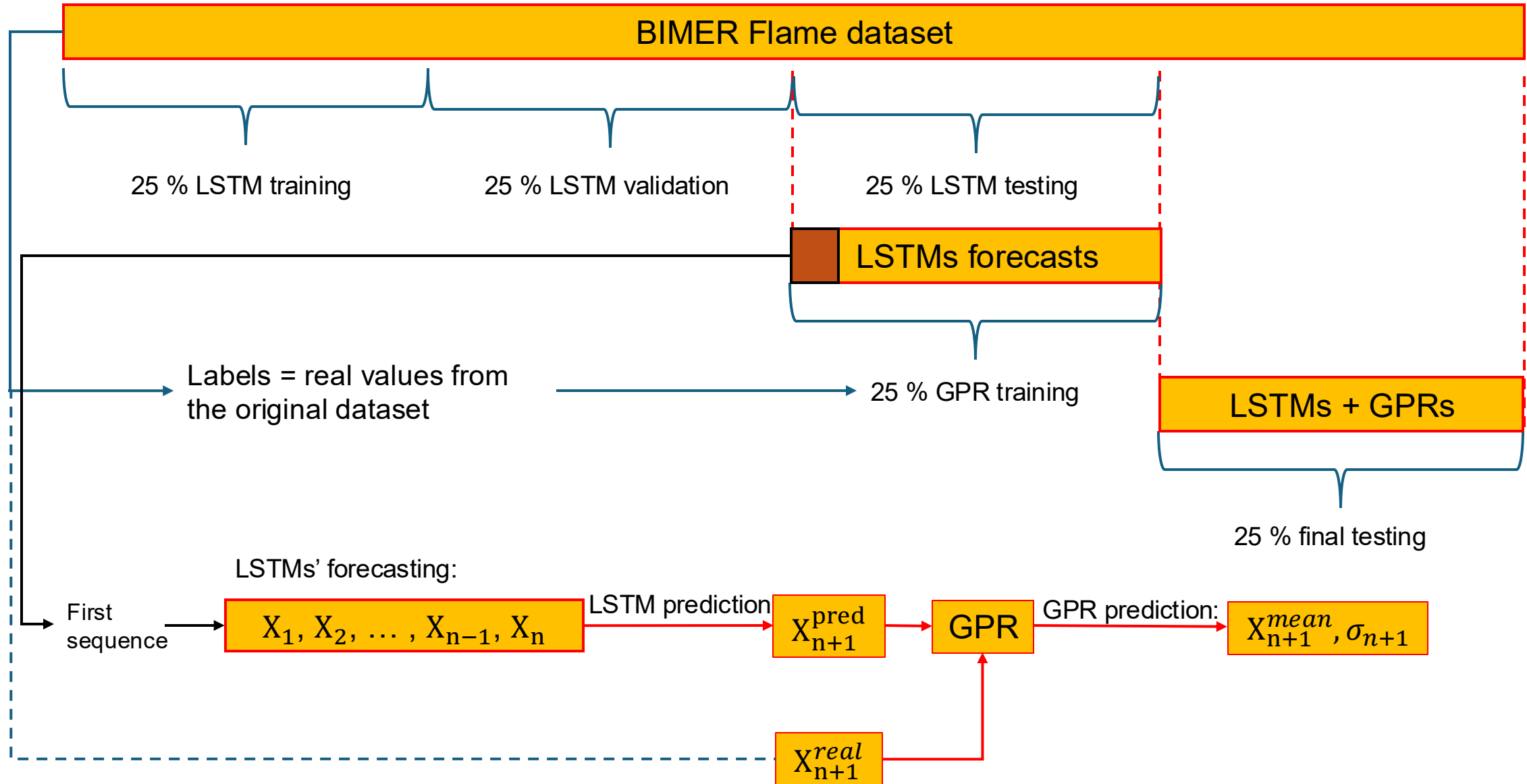
$$g(\gamma) \sim GP(\mu(\gamma), K(\gamma, \gamma'))$$

where $\mu(\gamma)$ is the **mean** function and $K(\gamma, \gamma')$ is a **covariance** function

- Prior: $\mathcal{P}(g) = \mathcal{N}(0, K(\gamma))$
- Likelihood: $\mathcal{P}(\mathbf{a}|g) = \mathcal{N}(0, \sigma_n^2 \mathbf{I})$
- Evidence: $\mathcal{P}(\mathbf{a})$
- Posterior: $\mathcal{P}(g|\mathbf{a}) = \frac{\mathcal{P}(\mathbf{a}|g)\mathcal{P}(g)}{\mathcal{P}(\mathbf{a})}$



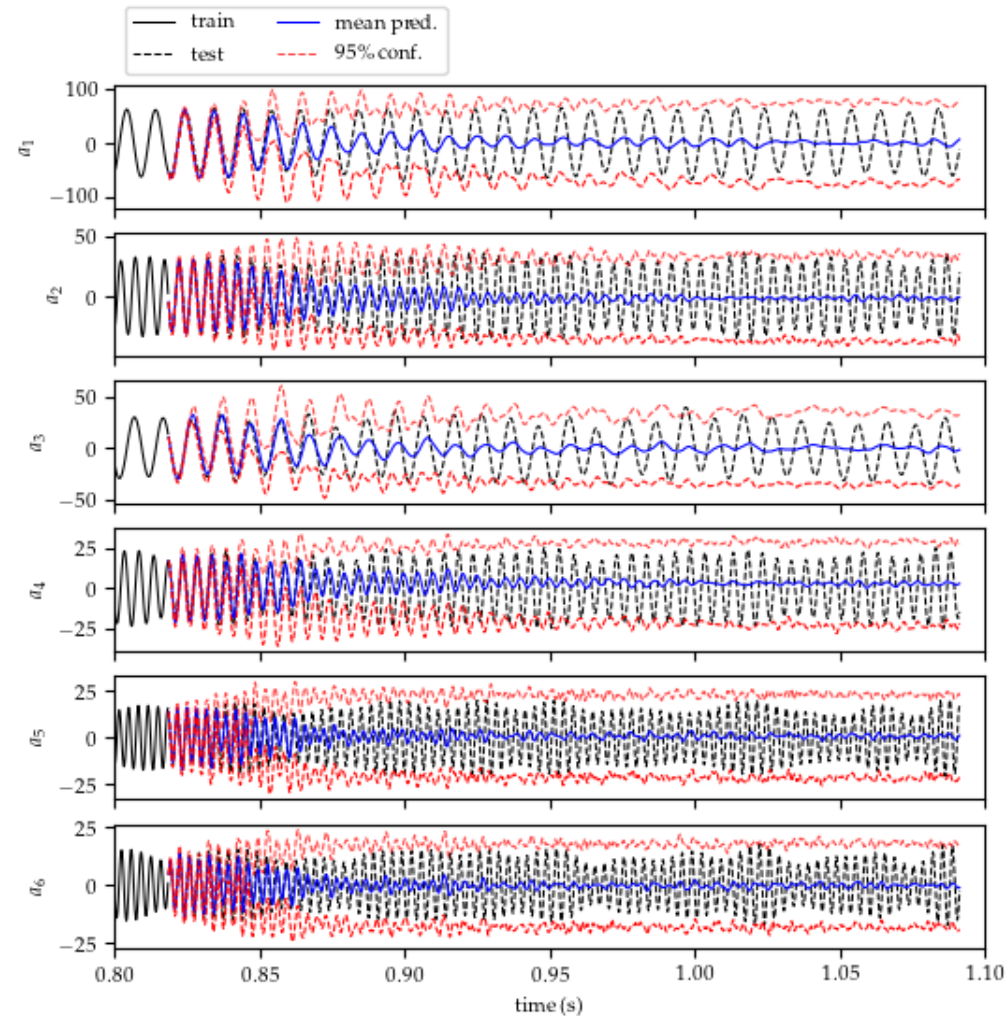
GPR Training



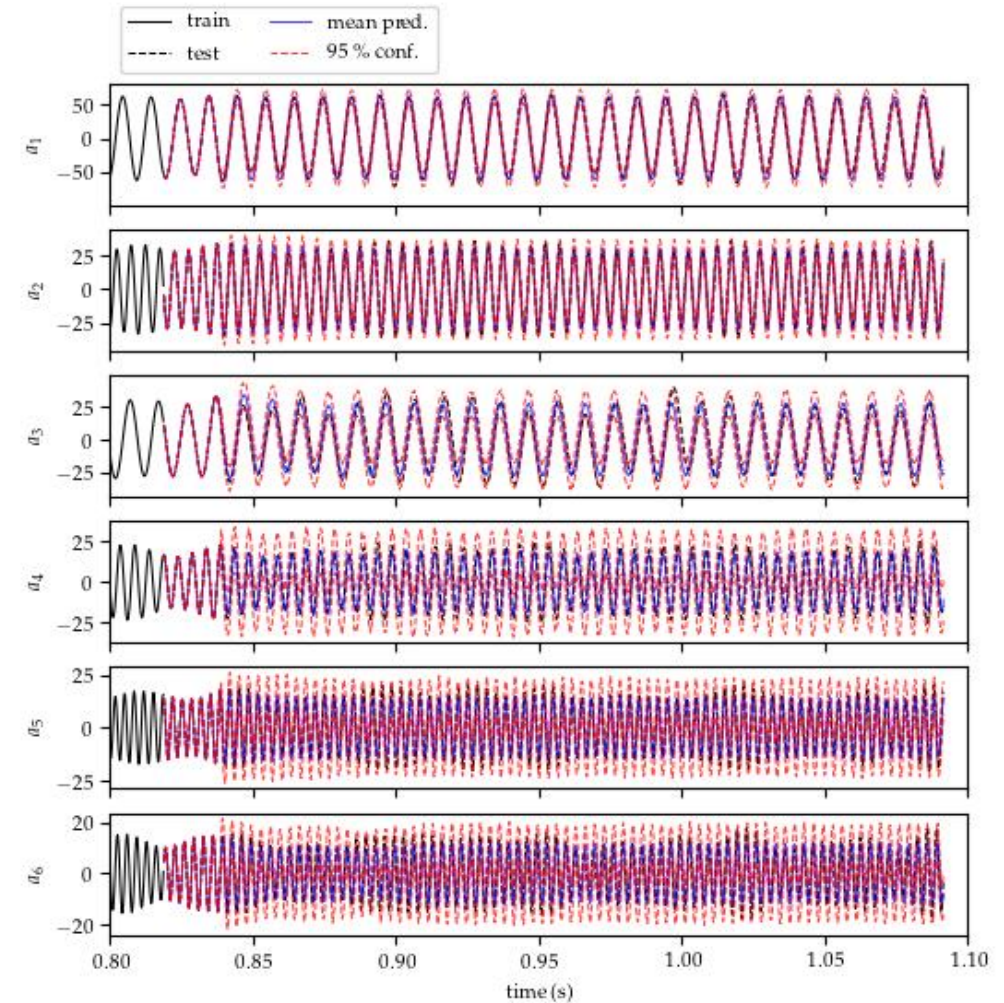
ROM forecasts provide stable long-term predictions

Compared with the GPR, the combination LSTM – GPR provides stable long-term predictions with associated uncertainty estimations

GPR model forecasts



LSTM - GPR model forecasts



Numerical Performance

LSTM provides accurate predictions, losing performance only on the less informative modes.

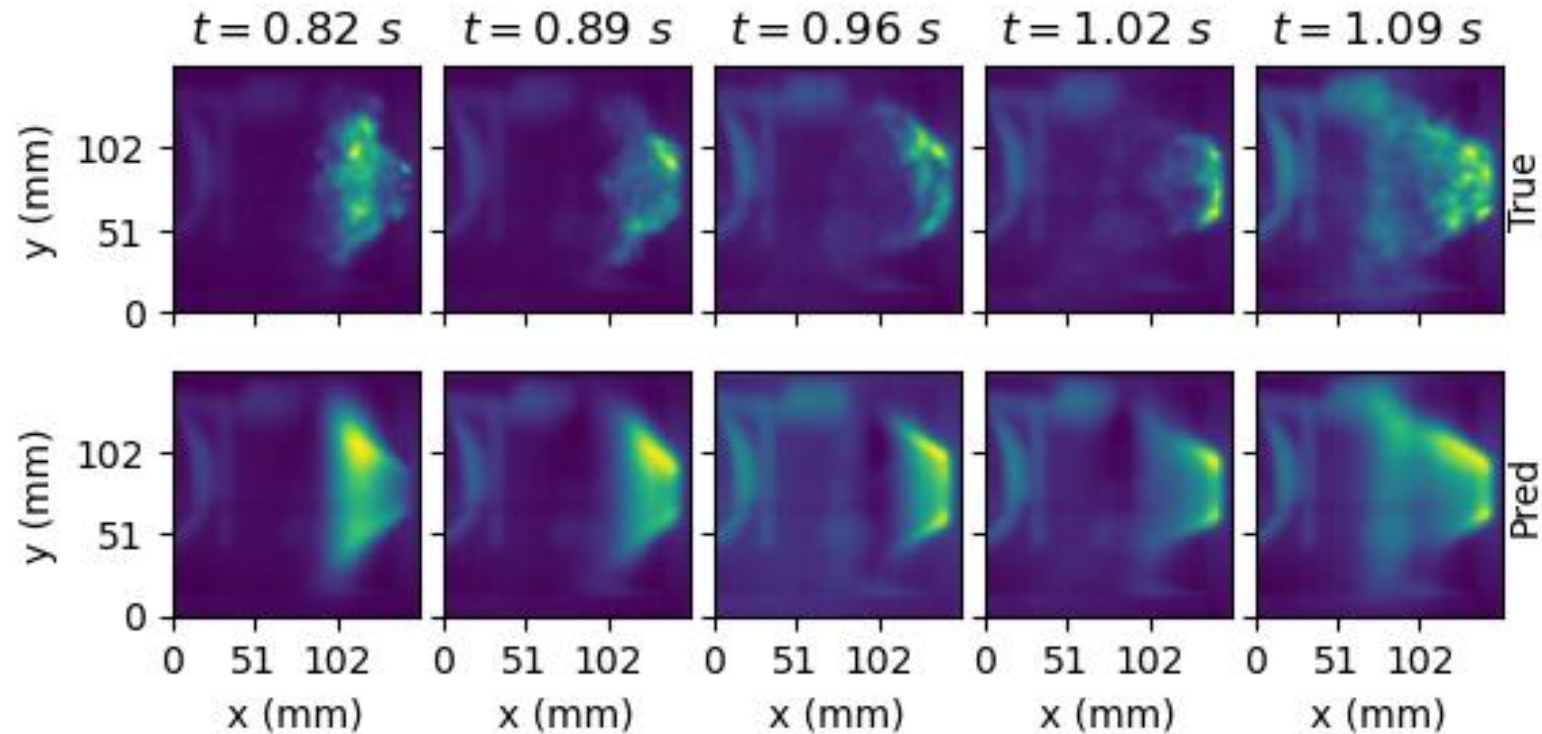
GPR contributes with:

- small improvements in accuracy for some modes;
- predictions' confidence interval of various amplitude according to the accuracy of each mode.

		Accuracy		Uncertainty			
Model	Modes	RMSE	R2	MPIW	PINAW	CI (%)	sigma
6 - GPR	1	37.67	0.24	133.25	0.97	99.37	33.31
	2	21.24	0.05	63.76	0.89	91.61	15.94
	3	18.56	0.21	63.62	0.84	94.86	15.90
	4	14.75	-0.02	46.37	0.93	92.48	11.59
	5	11.61	-0.02	42.50	1.05	99.53	10.63
	6	9.32	-0.01	34.72	0.99	99.68	8.68
6 LSTM+GPR	1	7.01	0.974	26.34	0.19	93.91	6.58
	2	3.80	0.970	14.37	0.20	94.22	3.59
	3	7.32	0.877	18.66	0.25	80.54	4.66
	4	3.85	0.931	26.32	0.53	100.0	6.58
	5	2.85	0.938	17.54	0.43	100.0	4.38
	6	5.42	0.659	14.79	0.42	81.41	3.70
Legend		• MPIW = Mean Prediction Interval Width • PINAW = Prediction Interval Normalized Average Width (normalized with range of signal, min-max) • CI = % of values contained in the Confidence Interval (width = 4 sigma)					

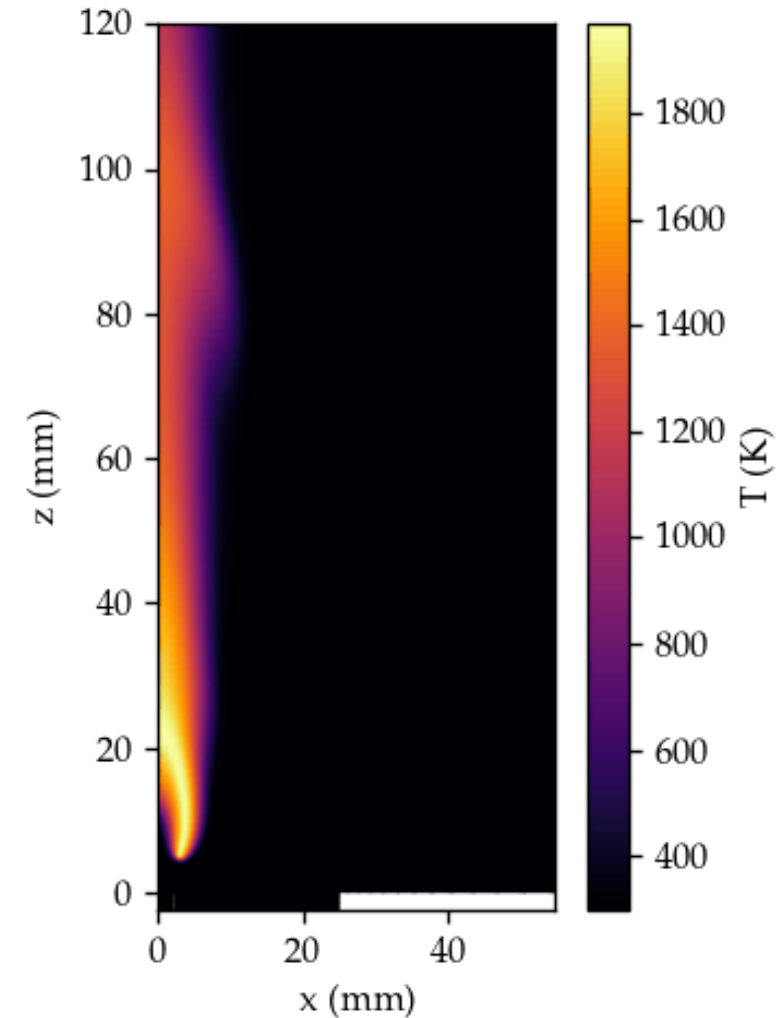
Reconstruction of the physical field

Original vs ROM predictions	
RMSE	8.411
NRMSE	3.89 %
rel. L2	13.94 %



mPOD – LSTM – GPR applied to a pulsated flame

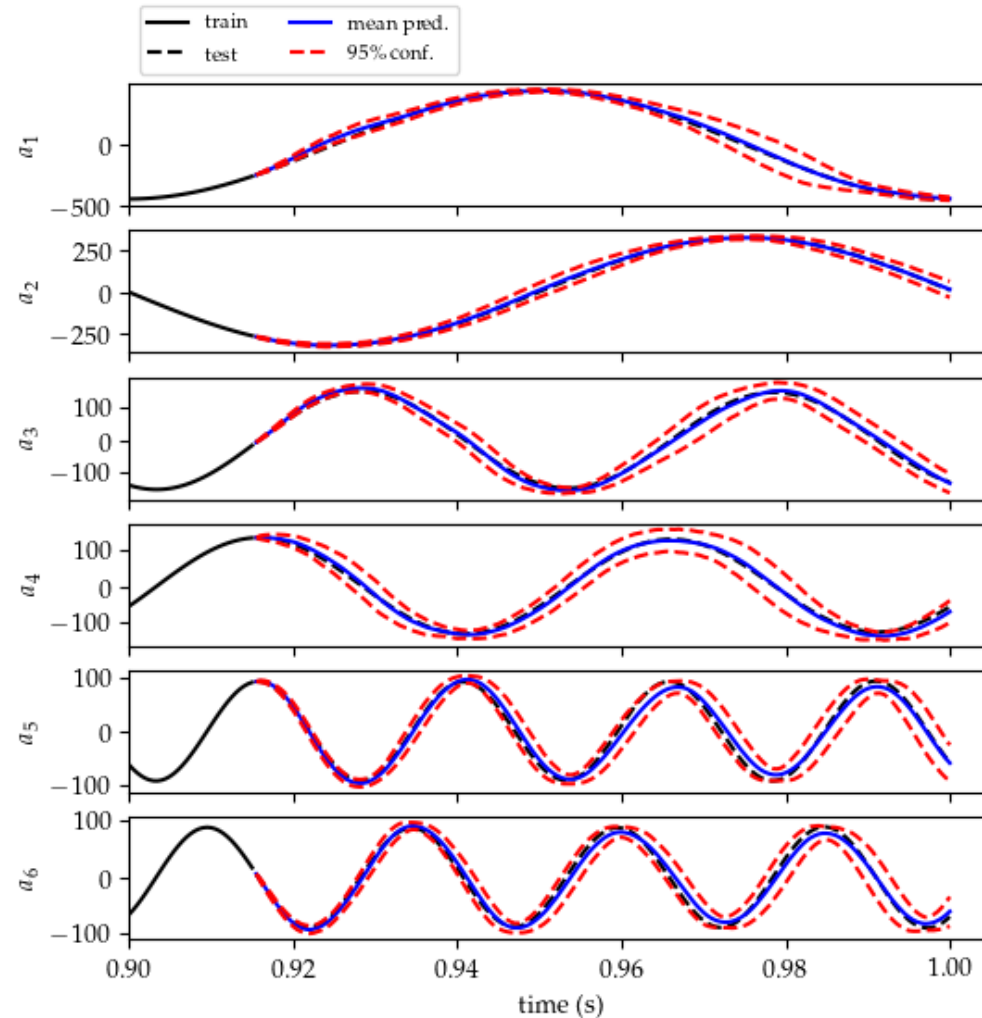
- 2D axisymmetric OpenFOAM simulation [1].
- Nitrogen-diluted methane-air flame.
- The forcing frequency is a combination of 10/20/80 Hz tones.
- The features included are:
 T , Y_{CH_4} , Y_{CO} , Y_{CO_2} , Y_{H_2} , Y_{H_2O} , Y_{N_2} , Y_{O_2} , Y_{OH} .
- The sampling rate is 2000 Hz.
- The number of snapshots is 2000.



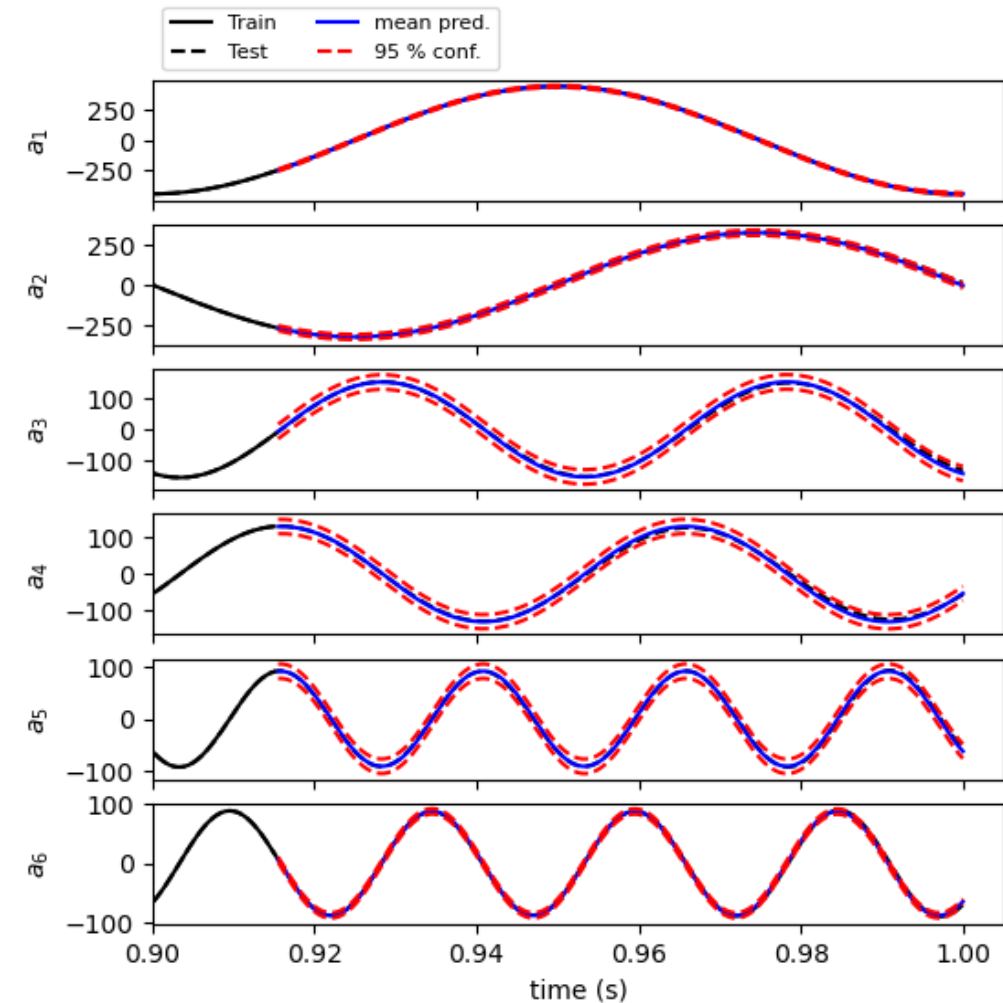
ROM forecasts provide stable long-term predictions

Compare with the GPR, the combination LSTM – GPR provides stable long-term predictions with associated smaller uncertainty estimations

GPR model forecasts



LSTM - GPR model forecasts



Numerical Performance

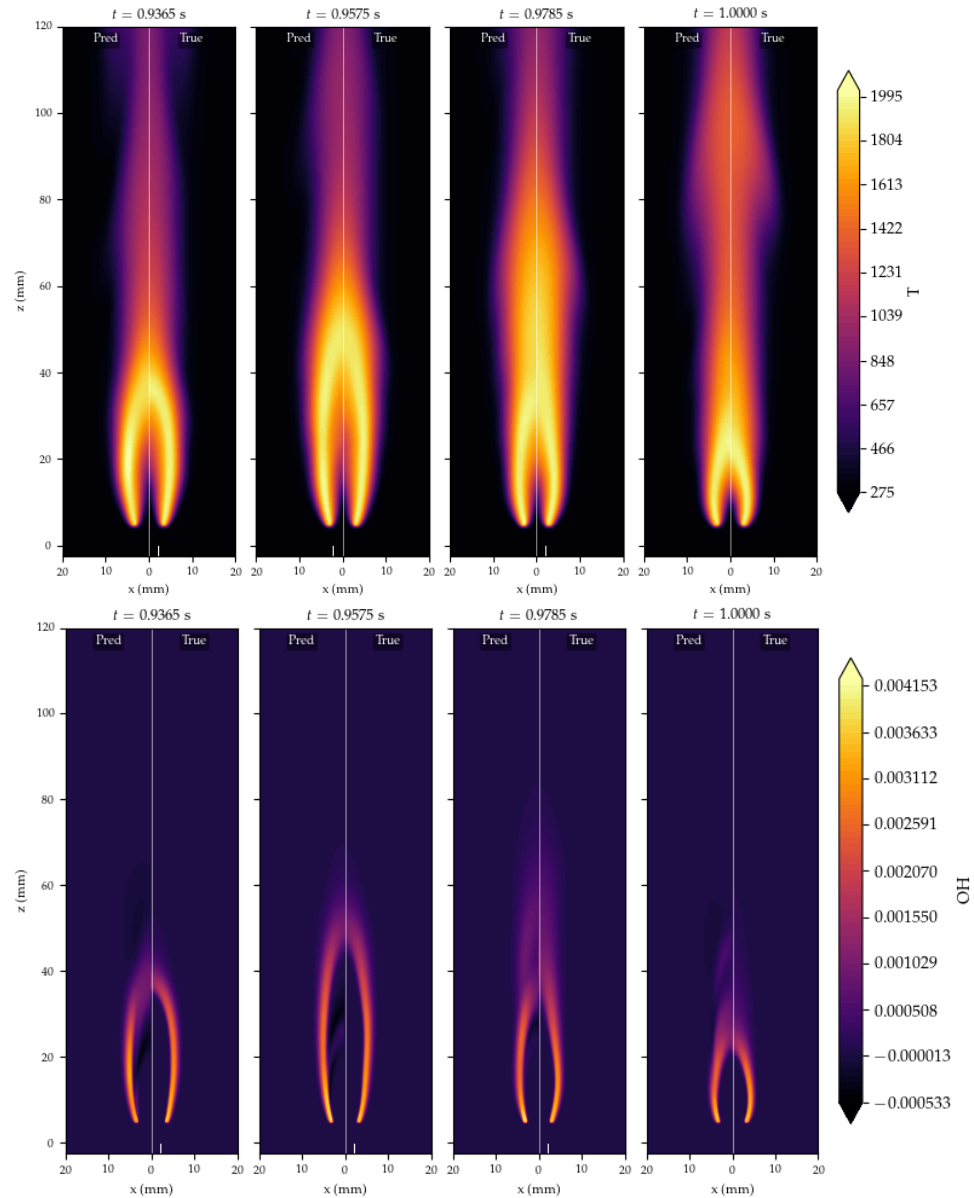
LSTM provides accurate predictions, losing performance only on the less informative modes.

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- predictions' confidence interval of various amplitude according to the accuracy of each mode.

		Accuracy		Uncertainty			
Model	Modes	RMSE	R2	MPIW	PINAW	CI (%)	sigma
6 - GPR	1	7.11	0.9994	73.03	0.082	100.00	18.26
	2	4.63	0.9996	75.01	0.116	99.41	18.75
	3	6.74	0.9957	46.15	0.152	100.00	11.54
	4	3.48	0.9986	36.03	0.138	100.00	9.01
	5	3.97	0.9962	31.71	0.170	99.41	7.93
	6	2.92	0.9978	29.67	0.166	100.00	7.42
6 LSTM+GPR	1	3.08	0.9999	16.74	0.019	100.00	4.19
	2	5.57	0.9995	32.51	0.050	100.00	8.13
	3	4.41	0.9981	46.47	0.153	100.00	11.62
	4	4.26	0.9979	38.83	0.149	100.00	9.71
	5	1.87	0.9992	27.79	0.149	100.00	6.95
	6	2.03	0.9990	9.03	0.050	92.94	2.26
Legend		• MPIW = Mean Prediction Interval Width • PINAW = Prediction Interval Normalized Average Width (normalized with range of signal, min-max) • CI = % of values contained in the Confidence Interval (width = 4 sigma)					

Reconstruction of the physical field



Feature	RMSE	NRMSE (%)	rel. L2 (%)
T	22.93	1.37	2.33
CH ₄	0.005205	1.01	6.68
CO	0.00133	2.71	10.22
CO ₂	0.002167	1.37	3.66
H ₂	5.111e-05	2.10	7.84
H ₂ O	0.001167	1.09	2.46
N ₂	0.00268	0.95	0.36
O ₂	0.002868	1.24	1.58
OH	0.0001174	3.09	20.62

Final Considerations and Future steps



Conclusive Analysis

The mPOD - LSTM - GPR pipeline is a valid combination to build a Bayesian Reduced Order Model for reacting flows.

- mPOD reduces the complexity of the problem;
- LSTM is able to accurately learn the signal behavior of the temporal modes of the system;
- GPR requires few hyperparameters to train, and is a solid uncertainty estimator;
- this pipeline is adaptable to different dynamical systems of reacting flows.

Future Works

- Applying the same combination of models to different cases: different multi-feature flame simulations and experimental flames.
- Explore **assimilation techniques** compatible with both deterministic nature of the LSTM networks and the stochastic nature of the GPR, for real-time state estimation of reacting flow: EnKF and bias-aware EnKF.
- Study and application of the **Deep Kernel Learning**[1] : it consists of inserting an LSTM network as a kernel function of the GPR.

Thank you for your attention!

Questions?

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